Ian Sequeira

Santa Barbara, CA | ian_sequeira@ucsb.edu | ian-sequ7.github.io/ian_sequeira |
linkedin.com/in/ian-sequ7 | github.com/ian-sequ7

Research Interests: anytime-valid inference, e-value theory, compositional reliability for multi-agent LLM systems

Education

University of California, Santa Barbara — Santa Barbara, CA
B.S. Statistics & Data Science, B.A. Economics — *Sept 2023 – June 2027*

- GPA: 3.83 / 4.0 | Dean's Honors: 7 quarters | Graduate coursework: PSTAT 231 Statistical Machine Learning (A)
- **Selected coursework:** Design of Experiments (A+), Regression Analysis (A), Stochastic Processes, Data Science (A), Machine Learning, Econometrics (A), Probability & Mathematical Statistics, Proof Writing, Data Structures

Research Experience

UCSB Department of Statistics & Applied Probability — Santa Barbara, CA
Undergraduate Researcher (PSTAT 199) – Prof. Annie Qu — *Apr 2026 – Present*

- Developing **Compositional Coordination Confidence (CCC)**, an anytime-valid runtime monitor for multi-agent LLM pipelines that attaches to any per-turn LLM-judge score and supplies a formal type-I guarantee via Ville's inequality on an e-process; guarantee holds at any stopping time, so operators may select α after observing the evidence
- Diagnosed and corrected degeneracy of the standard e-value construction on rubric-anchored LLM-judge output (five-anchor support, ceiling-heavy) that inflated measured false-alarm rate to $4\times$ nominal; introduced a categorical e-value construction that satisfies the calibration condition by construction rather than by estimation
- Role-conditional null with a tower argument establishing that per-role calibration preserves the supermartingale property; uniform location mixture over anchored wealth processes matched to the observed burst structure of coordination failures
- On held-out trajectories from two benchmarks, controls type-I error at nominal level under the plain Ville threshold (0.0% and 3.5% against $\alpha = 0.05$) with detection rates of 8.1% and 16.2%; on long chains, early halting saves $\sim$10% of per-trace tokens. Manuscript in preparation.

Summer Institute for Biostatistics & Data Science (iSI-BUDS) — UC Irvine, CA
Research Fellow – Prof. Dan Gillen & Prof. Josh Grill — *June 2025 – Aug 2025*

- Constructed GLM, ridge, and CART classifiers in R to identify Alzheimer's biomarkers (pTau217, PACC, ADL) across 1,150+ A4 preclinical trial participants; best model ROC-AUC 0.835
- Developed biomarker-guided sample-size framework achieving 80% statistical power with 188 participants vs. standard 500+
- Contributing co-author on the resulting biomarker-guided trial design study; results appeared as a co-authored poster at *CTAD 2025*, San Diego (see Publications)

Publications and Presentations

Sequeira, I., Qu, A. *Compositional Coordination Confidence for multi-agent LLM systems.* Manuscript in preparation. — 2026

Gaona-Partida, P., Anuurad, T., **Sequeira, I.**, Ragaine, E., Turner, C., Grill, J. D., Gillen, D. L. *Instructing Efficient Early-phase Preclinical Alzheimer's Disease Trials.* Manuscript in preparation. Preliminary results presented as a poster at *Clinical Trials on Alzheimer's Disease (CTAD)*, San Diego, CA, Dec 2025. — 2025 – 2026

Honors and Awards

SIOP Summer Research Scholarship, UCSB Statistics & Applied Probability (mentor: Prof. Annie Qu) — 2026

Hispanic Scholarship Fund (HSF) Scholar (renewable) — 2024 – Present

Dean's Honors, UC Santa Barbara (7 quarters) — 2023 – 2026

Professional Experience

Alida Inc. Remote

Data Engineering Intern *Sept 2024 – Present*

- Architected a multi-mode AI data exploration platform on Databricks operating over 47 healthcare datasets, combining multi-agent routing, graph analysis, and automated SQL generation
- Extended the platform with a spec-driven ML prediction pipeline featuring typed contracts, dual-layer verification, and fault-tolerant LLM orchestration with AutoML training

Turing Remote

Applied AI/GenAI Software Engineering Intern *Feb 2026 – May 2026*

- Delivered full SDLC on a 13-node LangGraph call-triage pipeline (Intake Agent) during a 12-week industry sprint, sourced through the ACM Industry program at UCSB; work combined retrieval-augmented routing with typed agent contracts

Teaching Experience

Workshop Instructor, ACM at UCSB 2025 – 2026

- Led student workshops on applied machine learning and agentic LLM systems for undergraduate ACM members

Selected Projects

Healthcare Financial Toxicity Prediction

PSTAT 231 (Graduate Statistical Machine Learning) final project *Mar 2026*

- Predicted catastrophic out-of-pocket burden in 20,962 MEPS 2022 respondents; tuned 6 classifiers under 10-fold cross-validation, XGBoost reaching ROC-AUC 0.874
- Evaluated subgroup fairness across race/ethnicity using DALEX and `fairmodels`; interpreted feature contributions as SHAP odds ratios to surface actionable clinical predictors

Research Advisor Finder

Full-stack faculty discovery platform (FastAPI, PostgreSQL/pgvector, Next.js) *Nov 2025 – Feb 2026*

- Matched student research interests to 15,000+ professors across 12 STEM fields via hybrid semantic search over CV-extracted interest embeddings

Leadership and Service

Project Executive, ACM Industry, UCSB Sept 2025 – Present

- Leading cross-functional student teams delivering applied data science and software solutions for industry partners including Turing

Finance & Funding Chair, CampMed, UCSB Sept 2025 – Present

- Leading budget planning and sponsorship strategy for health-equity outreach to underrepresented high school students

Technical Skills

Programming: R, Python, SQL, C++, TypeScript, LaTeX | **Methods:** e-processes, anytime-valid inference, GLMs, XGBoost, CART, SHAP, DALEX, Monte Carlo | **Tools:** tidymodels, rstan, scikit-learn, PyTorch, Databricks, LangGraph

Professional Memberships

American Statistical Association (ASA) | Institute of Mathematical Statistics (IMS) | UCSB Innovators Circle

References

Prof. Dan Gillen, Chair, Department of Statistics, UC Irvine | **Prof. Annie Qu**, Founding Director, Center for Statistical Foundations of AI, UCSB | **Jay Chyung, MD, PhD**, COO, Alida Inc.
(contact information available on request)